

## Normal Form Games · 1 topic

**Game:**  $(N, A, u)$ : agents,  $A = A_1 \times \dots \times A_n$ , payoffs  $u_i: A \rightarrow \mathbb{R}$ . 2 players + finite  $\Rightarrow$  matrix (row=P1, col=P2). **Best response:**  $a_i \in \text{BR}(a_{-i})$  iff  $u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i}) \forall a'_i$ . **Strictly dominated:**  $\exists a'_i: u_i(a'_i, a_{-i}) > u_i(a_i, a_{-i})$  for every  $a_{-i}$ .

**IESDS** repeatedly delete strictly dominated actions; survivors = prediction.

**STRICT** only. Deleting **weakly** dominated can kill real NE + is order-dependent.

**Pure NE** nobody gains by unilateral deviation. **Find:** mark each player's best-response payoff per opponent action; cell with all marks = NE (can be 0, 1, many). **Pareto optimal:** no other outcome makes everyone  $\geq$  as well off AND someone strictly better. (Coordination/BoS pure NE are PO; PD's (D,D) is NOT.) **Story  $\rightarrow$  game:** list  $N$  (players),  $A_i$  (each one's options), fill  $u$  from text: "want to match"  $\Rightarrow$  match cells +, mismatch 0; "each prefers own option"  $\Rightarrow$  asymmetric  $\frac{2,1}{1,2}$ .

**Mixed NE — indifference (2×2)**

1. Opponent plays an action w.p.  $p$ .
2. Write your  $\text{EU}(\text{act}_1)$ ,  $\text{EU}(\text{act}_2)$  as fns of  $p$ .
3. Set **equal**  $\rightarrow$  solve  $p$ . Repeat for other player.

The  $p$  making ME indifferent = prob. the OPPONENT plays. Every finite game has  $\geq 1$  NE (maybe mixed).

**Worked — Battle of Sexes** (the recurring "model + solve" game). Max=row, Ann=col;  $II = (1, 2)$ ,  $IT = (0, 0)$ ,  $TI = (0, 0)$ ,  $TT = (2, 1)$ . **Pure:** underline BRs  $\Rightarrow (I, I)$ ,  $(T, T)$ , both Pareto optimal. **Mixed:** Ann plays  $I$  w.p.  $q$ , Max indiff:  $1q + 0 = 0 + 2(1 - q) \Rightarrow q = \frac{2}{3}$ . Max plays  $I$  w.p.  $r$ , Ann indiff:  $2r + 0 = 0 + 1(1 - r) \Rightarrow r = \frac{1}{3}$ . Mixed NE  $(\frac{2}{3}, \frac{1}{3})$ ,  $(\frac{1}{3}, \frac{2}{3})$ .

**Canonical games**

- **Prisoner's Dilemma:** dominance  $\Rightarrow$  unique bad NE (defect, defect).
- **Matching Pennies** (zero-sum): no pure NE; mixed  $(\frac{1}{2}, \frac{1}{2})$ .
- **Battle of Sexes:** 2 pure NE + 1 mixed; players mix with different probs.

**Cournot duopoly**  $P(Q) = a - Q$ ,  $Q = q_1 + q_2$ , cost  $c$ .  $\pi_i = q_i(a - q_1 - q_2 - c)$ . FOC  $\Rightarrow b_i(q_j) = \frac{1}{2}(a - q_j - c_i)$ .

Symmetric NE:  $q_1^* = q_2^* = \frac{1}{3}(a - c)$ . **Asymmetric costs:** solve BOTH reaction fns simultaneously (don't assume  $q_1 = q_2$ ). Demand  $P = a - bQ$ :  $b_i = \frac{a - c_i - bq_i}{2b}$ . Plug back to check.

## Extensive Form Games · 2

Tree = sequential moves. Node=choice, edge=action, leaf=payoff vector. **Strategy: complete contingent plan:** an action at every own node / info set, even unreached ones.

Unreached-node actions  $\Rightarrow$  normal-form redundancy + let NE rest on **non-credible threats**.

**Ext  $\rightarrow$  Normal** rows/cols = full strategy sets; cell = leaf reached; find NE normally. Round-trip need not redraw the same tree but games are **strategically equivalent** (same equilibria).

**Imperfect info:** nodes the mover can't tell apart joined in one info set (dashed line).

Simultaneous matrix game  $\Rightarrow$  single info set.

**SPE NE** in every subgame. Rules out non-credible threats. Every SPE is a NE (not vice-versa);  $\geq 1$  exists. **Disprove:** find one subgame where the restriction isn't a NE.

**Backward induction (gives SPE)** leaves up: at node  $h$ , mover picks child maximizing **their own** payoff coordinate; propagate the whole vector up.

Compare on the **moving** player's component only. Zero-sum: BI = minimax (+ alpha-beta).

## Bayesian Games · 3

**Game:**  $(N, A, \Theta, p, u)$ : types  $\Theta_i$ , prior  $p(\theta)$ , payoffs  $u_i(a, \theta)$  depend on types. **BNE:** each type best-responds in **expectation** over opponents' types:

$$\max_{a_i} \sum_{\theta_{-i}} p(\theta_{-i} | \theta_i) u_i(a_i, a_{-i}(\theta_{-i}), \theta)$$

**Compute** per own type, expected utility of each action over opponent types (conditional prob.), pick best. Check no type deviates. Strict dominance checkable **per type** (often pins one player first).

**Solve-for-p:** write  $\text{EU}(a_1 | \text{type})$ ,  $\text{EU}(a_2 | \text{type})$  as linear fns of prior  $p$ ; solve  $\text{EU}(a_2) > \text{EU}(a_1)$  for  $p$ -interval. An agent that KNOWS its type  $\Rightarrow$  its BR is per-type, does NOT depend on  $p$ .

## Repeated Games · 4

**Discount**  $\beta \in (0, 1)$ : weight on future;  $\sum \beta^t r_t$ . Higher = more patient.

Geom:  $\sum_{t=0}^{\infty} \beta^t = \frac{1}{1-\beta}$ . Constant  $c$  from now =  $\frac{c}{1-\beta}$ ; from next period =  $\beta \frac{c}{1-\beta}$ .

**As a tree:** draw stage game once per period; a path's payoff = SUM of stage payoffs (discounted by  $\beta^t$  if asked).

- **Finite:** backward induction  $\Rightarrow$  play stage-NE every round (cooperation unravels).
- **Infinite:** cooperation sustainable via threats.

**Grim trigger (infinite PD)** cooperate until a defection, then defect forever. NE iff patient enough:

$\beta \geq \frac{T-R}{T-P}$  ( $T$ =temptation,  $R$ =reward,  $P$ =punish): future punishment loss  $\geq$  one-shot defect gain.

**Folk thm:** any feasible payoff  $>$  minimax is a NE for  $\beta$  near 1.

## Multiagent Learning · 5 (12%!)

**Fictitious play** belief = COUNT vector of opp's past actions (init counts given).

1.  $\text{EU}(\text{act}) = \sum_{\text{opp}} \left( \frac{\text{count}_{\text{opp}}}{\text{total}} \right) \cdot u(\text{act}, \text{opp})$ .
2. Play arg max; THEN +1 to the action opp actually played. Ties: stated rule.

Init counts: magnitude = confidence, ratio = prior freq. If it converges  $\Rightarrow$  that point is a NE. Can cycle (Shapley).

**Q-learning / Nash-Q**

- **Indep** learners: own  $Q_i(s, a_i)$ , ignore others (non-stationary).
- **Joint-action** learners:  $Q_i(s, \vec{a})$ . Table size: indep =  $|A_i|$ ; joint/central =  $|A_1| \cdot \dots \cdot |A_n|$ .
- **Matrix game** (no state,  $\gamma = 0$ ):  $Q_i[\vec{a}] \leftarrow Q_i + \alpha(r_i - Q_i)$ . Greedy picks NE of current  $Q$  (init 0  $\Rightarrow$  first step ties). Minimax-Q / Nash-Q replace max by game value.

**Markov game:**  $(N, S, \{A_i\}, P, \{R_i\}, \gamma)$ : MDP with joint actions + per-agent rewards; solve model-free w/ (minimax/Nash) Q-learning.

## Auctions · 6

**Second-price/Vickrey:** highest wins, pays 2nd price  $\Rightarrow$  **truthful bidding dominant**. **First-price sealed:** highest wins, pays own bid  $\Rightarrow$  **shade**. Sym. eq. ( $n$  bidders, iid  $U(0, 1)$ ):  $b(v) = \frac{n-1}{n} \cdot v$ .

**Payoff** = valuation - price paid (add commission to winner's cost). **Efficient** iff winner = highest-valuation agent. 2nd-price under uncertainty: bid the EXPECTED value of winning.

English (asc)  $\approx$  2nd-price; Dutch (desc)  $\approx$  1st-price.

## Agents & Coordination · 7

- **Behaviours:** reactive (timely) / proactive (goal) / social. Social = coordination (interdependencies) vs cooperation (shared goal) vs negotiation (conflicting interests). Don't conflate.
- **Subsumption:** behaviour layers, **lower inhibits higher**; no model, **no learning**; hard to add layers.
- **Deductive/logic agent cons:** theorem-proving costly; **calculative-rationality** (world changes mid-decision); hard to model dynamic worlds.
- **BDI:** practical reasoning = deliberation (WHAT goal) + means-ends (HOW = plan). Commitment: blind / single-minded / open-minded.
- **Hybrid:** horizontal layering (all layers see I/O, fault-tolerant, needs mediator) vs vertical (pipeline).
- **Rational agent:**  $a^* = \arg \max_a \sum_o u(o) P(o|a)$ ; spec via PEAS.
- **Coordination:** social convention = fixed agent+action ordering  $\Rightarrow$  deterministic, no conflict. Factor graph:  $\vec{a}^* = \arg \max \sum_j f_j$  via **variable elimination** (eliminate agents one by one).

**EXTRA — worked examples (copy top-down till full)**

### 1. Fictitious play (2025 exam)

game  $U/D \times L/R$ ,  $(P_1, P_2)$ :  $UL = (1, 2)$ ,  $UR = (5, 6)$ ,  $DL = (4, 6)$ ,  $DR = (7, 1)$ . ( $D$  dominates for P1; unique NE  $(D, L)$ .) Init counts  $(1, 1)$  each.

- R1: P1 belief  $(\frac{1}{2}, \frac{1}{2})$ :  $\text{EU}(U) = \frac{1}{2} + \frac{5}{2} = 3$ ,  $\text{EU}(D) = 2 + \frac{7}{2} = 5.5 \Rightarrow D$ . P2:  $\text{EU}(L) = 1 + 3 = 4$ ,  $\text{EU}(R) = 3 + \frac{1}{2} = 3.5 \Rightarrow L$ . Saw  $(D, L)$ .
- R2: P1 belief  $L = \frac{2}{3}$ :  $\text{EU}(U) = \frac{7}{3}$ ,  $\text{EU}(D) = 5 \Rightarrow D$ . P2  $\Rightarrow L$ .

Converges to the unique NE  $(D, L)$  — **regardless of initialization**.

### 2. Nash-Q joint-action (2025 exam)

same game,  $\gamma = 0$ . Current  $Q_R = [U : 1, 4; D : 3, 7]$ ,  $Q_C = [U : 2, 5; D : 5, 1]$ .

- Greedy = NE of current  $Q$ :  $(D, L)$  ( $Q_R = 3$ ,  $Q_C = 5$ ; deviating lowers both). Play it; observe  $r_R = 4$ ,  $r_C = 6$ .
- Update:  $Q_{R(D,L)} = 3 + \alpha(4 - 3) = 3 + \alpha$ ;  $Q_{C(D,L)} = 5 + \alpha(6 - 5) = 5 + \alpha$ . NE unchanged  $\Rightarrow$  converged. Sizes: indep =  $|A_i|$ ; joint =  $|A_1| |A_2|$ .

### 3. BNE — solve-for-p (2023 exam)

P1 type  $t_1$ ; P2 type  $A$  (w.p.  $p$ ) or  $B$ . If  $A$ :  $X$  dominates for P2; if  $B$ :  $Y$  dominates. So  $a_2^* = (X \text{ if } A, Y \text{ if } B)$ .

- P1:  $EU(X) = p \cdot 0 + (1 - p)2 = 2(1 - p)$ ;  
 $EU(Y) = p \cdot 3 + (1 - p)1 = 2p + 1$ .
- P1 plays  $Y$  iff  $2p + 1 > 2(1 - p) \Rightarrow p > \frac{1}{4}$ .

#### 4. Extensive form — 3 P1-nodes (2025 exam)

$P1\{A, B\}; A \rightarrow P2\{C, D\} \quad (C \rightarrow P1\{G, H\} : G(3, 8), H(8, 3); D \rightarrow (6, 6)); B \rightarrow P2\{E, F\} (E \rightarrow (4, 3); F \rightarrow P1\{I, J\} : I(4, 5), J(5, 7)).$

- **BI:** P1 leaves:  $H$  ( $8 > 3$ ),  $J$  ( $7 > 5$ ). P2 left:  $D$  ( $6 > 3$ )  $\rightarrow$  ( $6, 6$ ). P2 right:  $F$  ( $7 > 3$ )  $\rightarrow$  ( $5, 7$ ). Root:  $A$  ( $6 > 5$ ).

- **SPE:**  $\{(A, H, J), (D, F)\}$  (P1 strategy lists all 3 nodes).

#### 5. Grim trigger — $\beta$ threshold (2022 exam)

PD  $(C, C) = 6, 6$ ; defect  $T = 15$  once then punished  $P = 3$ ;  $R = 6$ .

- Coop forever  $\frac{6}{1-\beta}$  vs defect  $15 + \beta\left(\frac{3}{1-\beta}\right)$ .
- Diff  $-9 + \beta\left(\frac{3}{1-\beta}\right) \geq 0 \Rightarrow \beta \geq \frac{T-R}{T-P} = \frac{15-6}{15-3} = \frac{3}{4}$ . “Value tomorrow  $\geq \frac{3}{4}$  of today.”

#### 6. Cournot asymmetric (2023 exam)

$P = 100 - 2Q$ ,  $c_A = 20, c_B = 10$ .  $\frac{\pi_A}{-2q_A^2 - 2q_Aq_B + 80q_A}$ ; FOC  $q_A = \frac{40-q_B}{2}$ ,  $q_B = \frac{45-q_A}{2}$ . Solve  $\Rightarrow q_A = \frac{35}{3}, q_B = \frac{50}{3}, P = \frac{130}{3}$ .

#### 7. Pareto on PD

NE  $(D, D)$  is NOT Pareto optimal —  $(C, C)$  beats it for both. NE  $\neq$  Pareto.